

## Introduction

Simple linear regression models the mean of a continuous outcome as a linear function of a single continuous predictor. It is the workhorse of applied statistics: half of all reported analyses are linear regressions in some disguise, and the concepts it introduces (least squares estimation, residuals, fit-vs-data trade-off, R-squared, confidence and prediction bands) generalise to every more complex model.

## Prerequisites

The reader should know what a mean, a variance, and a correlation are, and should be comfortable reading a scatter plot. No prior regression experience is required.

## Theory

Given pairs  $(x_i, y_i)$  for  $i = 1, \dots, n$ , the simple linear regression model is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2).$$

The parameters  $\beta_0$  (intercept) and  $\beta_1$  (slope) are estimated by ordinary least squares – minimising the sum of squared residuals  $\sum (y_i - \hat{y}_i)^2$ . Closed-form solutions are

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The standard error of the slope is  $SE(\hat{\beta}_1) = s / \sqrt{\sum (x_i - \bar{x})^2}$ , where  $s$  is the residual standard error. Under the normality assumption,  $\hat{\beta}_1$  is normally distributed around  $\beta_1$ , and  $\hat{\beta}_1 / SE(\hat{\beta}_1)$  follows a  $t$  distribution with  $n - 2$  degrees of freedom under the null  $\beta_1 = 0$ .

The coefficient of determination is  $R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$  – the proportion of variance in  $y$  explained by the linear relationship with  $x$ . For simple regression,  $R^2$  equals the squared Pearson correlation.

## Assumptions

The OLS inference machinery relies on four assumptions:

1. **Linearity:** the true relationship between  $x$  and the mean of  $y$  is linear.
2. **Independence:** the errors  $\varepsilon_i$  are independent of each other.
3. **Homoscedasticity:** the errors have constant variance  $\sigma^2$  regardless of  $x$ .
4. **Normality of errors:** the errors are normally distributed – important for small-sample inference, less so for large samples.

## R Implementation

```
library(broom)
library(ggplot2)
library(performance)

set.seed(7)
n <- 80
x <- runif(n, 0, 10)
y <- 2 + 1.5 * x + rnorm(n, sd = 1.5)
df <- data.frame(x, y)

fit <- lm(y ~ x, data = df)
```

```

tidy(fit, conf.int = TRUE)
glance(fit)

ggplot(df, aes(x = x, y = y)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", colour = "steelblue") +
  theme_minimal(base_size = 12)

check_model(fit)

```

The `tidy()` function returns a one-row-per-term table with estimates, standard errors, *t* statistics, *p*-values, and confidence intervals. `glance()` returns a one-row model-level summary: *R*-squared, adjusted *R*-squared, residual standard error, *F* statistic, and log-likelihood. `check_model()` produces the standard diagnostic panel.

## Output & Results

Typical coefficient output:

| term        | estimate | std.error | statistic | p.value | conf.low | conf.high |
|-------------|----------|-----------|-----------|---------|----------|-----------|
| (Intercept) | 1.982    | 0.326     | 6.080     | <0.001  | 1.333    | 2.631     |
| x           | 1.512    | 0.056     | 27.155    | <0.001  | 1.401    | 1.623     |

$R^2 = 0.904$ , residual SE = 1.47,  $F(1, 78) = 737.4$ .

## Interpretation

The slope estimate of 1.512 (95% CI 1.401 to 1.623) says that a one-unit increase in  $x$  is associated with a 1.51-unit increase in the conditional mean of  $y$ , holding all else equal (vacuous for a simple regression, but the phrasing generalises). The intercept of 1.982 is the predicted  $y$  when  $x = 0$ ; this is only a scientifically meaningful quantity if  $x = 0$  is within the range of the data.

For a manuscript: “A simple linear regression showed that  $y$  increased significantly with  $x$ ,  $b = 1.51$  (95% CI 1.40 to 1.62),  $t(78) = 27.2$ ,  $p < 0.001$ . The model explained 90.4% of the variance in  $y$ .”

## Practical Tips

- Always plot the data and the fitted line; numbers alone can hide gross misfit.
- Residuals-vs-fitted and Q-Q plots are the two diagnostics you should reflexively inspect.
- An  $R^2$  of 0.9 is stellar in physics, mediocre in psychology, and suspect in epidemiology – interpret it in context.
- The confidence interval is for the *mean* response at a given  $x$ ; the prediction interval, which is wider, is for a *new individual* observation.
- Extrapolating beyond the range of the observed  $x$  is risky: the linear relationship that held for  $x \in [0, 10]$  may be nonsense for  $x = 100$ .

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.

- **What an adequate Methods paragraph must contain.**

**See also — labs in this chapter**

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI